Dynamic Labor Demand and Adjustment Costs

Labor II — Week 2

Teaching deck draft

Central question

- Why do firms adjust employment slowly after shocks?
- Which cost structures generate smooth responses versus inaction and lumpiness?
- Why do policy effects depend on the adjustment path rather than only the new target?

Week 1 to Week 2 bridge

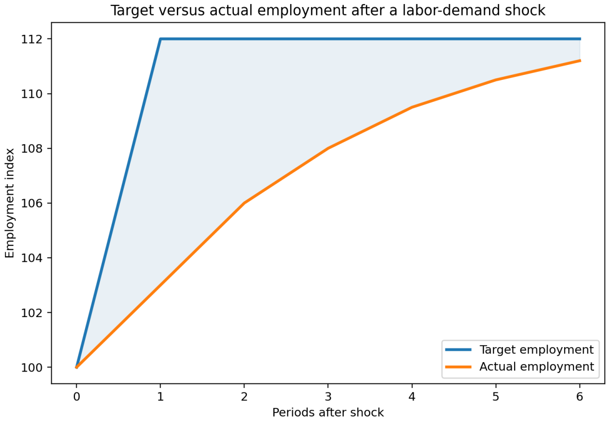
- Week 1 derived the firm's static labor-demand target.
- Week 2 asks why actual employment does not jump immediately to that target.
- Week 3 will stay inside the firm and ask how personnel design changes internal adjustment.

Target versus actual employment

$$L_t^* = L^*(w_t, p_t, K_t, A_t, q_t)$$

- The target is the Week 1 destination.
- The Week 2 object is the gap between actual employment and that target.
- The horizon matters: hours, headcount, hiring, and firing need not move together.

Target versus actual after a shock



Dynamic objective with adjustment costs

$$V(L_{t-1}, s_t) = \max_{L_t} \{\pi(L_t, s_t) - C(L_t - L_{t-1}) \; + \; \beta E_t[V(L_t, s_{t+1})]\}$$

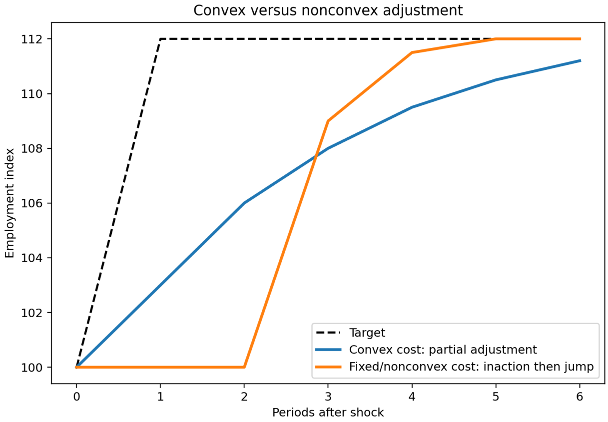
- Yesterday's workforce is a state variable.
- Adjustment costs make current choices affect future payoffs.
- The cost function $C(\cdot)$ is the central Week 2 object.

Convex costs and partial adjustment

$$L_t - L_{t-1} = \lambda(L_t^* - L_{t-1}), \qquad 0 < \lambda \leq 1$$

- Convex costs imply frequent but modest adjustments.
- λ is a reduced-form speed-of-adjustment object.
- Short-run incidence can differ sharply from long-run incidence.

Convex versus nonconvex adjustment



Fixed or nonconvex costs

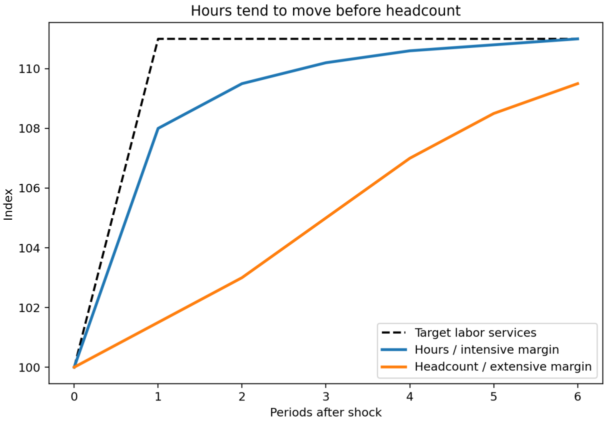
$$\text{adjust if } |L_t^* - L_{t-1}| > \bar{\Delta}$$

- Small shocks generate inaction.
- Large shocks trigger lumpy adjustments.
- Plant-level zeros and spikes are hard to reconcile with a purely convex benchmark.

Hours, headcount, hiring, and firing

- Hours often move before core headcount.
- Hiring margins are most visible after positive shocks.
- Separations and layoffs are most visible after negative shocks.
- Labor hoarding means weak headcount responses can still mask real labor-demand adjustment.

Hours versus headcount



Empirical designs and what they identify

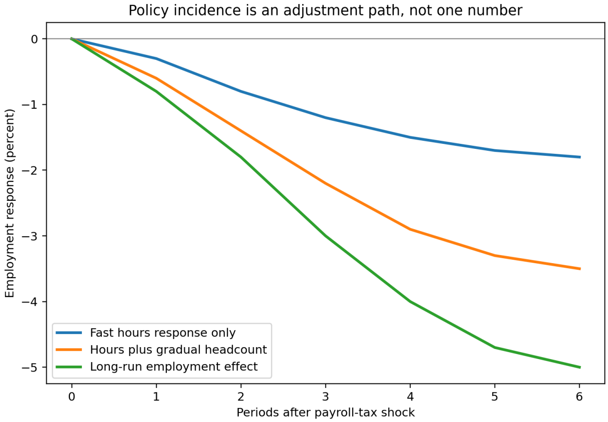
Design	Variation	Margin	Dynamic object
Plant panels	Within-firm dynamics	Employment, hours	Persistence, hazards, lumpiness
Policy timing	Labor-cost reform	Wages, employment, hours	Incidence over event time
Belief or uncertainty surveys	Firm expectations	Expected or realized employment	Delay and state dependence
Structural estimation	Model-implied states	Adjustment path	Deep cost parameters

Policy timing and incidence

$$\frac{\partial L_{t+h}}{\partial \tau_t}, \qquad h = 0, 1, 2, \ldots$$

- Dynamic policy effects are horizon-specific.
- Firms can adjust hours, vacancies, wages, prices, or headcount in sequence.
- Event studies and structural models answer related but different questions.

Policy incidence over time



Bridge to Week 3

- Week 2 explains why employment paths are slow and margin-specific.
- Week 3 asks how firms design incentives, promotions, and internal labor markets once those dynamic frictions exist.
- Dynamic adjustment is therefore the firm-side foundation for the personnel block.

Takeaways

- Target employment and actual employment are different objects.
- Convex costs smooth responses; fixed or nonconvex costs create inaction and lumpiness.
- Hours, headcount, hiring, and firing reveal different adjustment margins.
- Policy incidence should be read over the adjustment path, not at one horizon only.